

JAVA – DATABASE CONNECTION

(JDBC)

TYPE 1:

```
package JDBC;
import java.sql.*;
public class Type1Eg {
    public static void main(String [] args) throws Exception{
        readRecords();
        insertRecords();
        insertVar();
        insertUsingPst();
        update();
        commitdemo();
    }
    public static void readRecords() throws Exception{
        String url = "jdbc:mysql://localhost:3306/jdbc";
        String username = "root";
        String password ="Shalu7Selvaraj";
        String query = "Select * from employee";
        Connection con =
DriverManager.getConnection(url,username,password);
        System.out.println("Connected to DataBase Successfully");
        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery(query);
```

```

        while (rs.next()) {
            System.out.println("Id : "+rs.getInt(1));
            System.out.println("Name : "+rs.getString(2));
            System.out.println("Salary : "+rs.getInt(3));

        }
        con.close();
    }

    public static void insertRecords() throws Exception{
        String url = "jdbc:mysql://localhost:3306/jdbc";
        String username = "root";
        String password = "Shalu7Selvaraj";
        String query = "insert into employee values
(1,'Priya',100000)";

        Connection con =
DriverManager.getConnection(url,username,password);

        Statement st = con.createStatement();
        int rows = st.executeUpdate(query);
        System.out.println("Number of rows affected:"+rows);
        st.close();
        con.close();
    }

    public static void insertVar() throws Exception{
        String url= "jdbc:mysql://localhost:3306/jdbc";
        String username = "root";
        String password = "Shalu7Selvaraj";

```

```
        int id = 2;
        String name = "Varun";
        int salary = 250000;
        String query = "insert into employee values (" + id + ", '" +
name + "', " + salary + ")";
        Connection con =
DriverManager.getConnection(url,username,password);
        Statement st = con.createStatement();
        int rows = st.executeUpdate(query);
        System.out.println("Number of rows affected: " + rows);
        con.close();
```

```
}
```

```
public static void insertUsingPst() throws Exception{
    String url = "jdbc:mysql://localhost:3306/jdbc";
    String username = "root";
    String password = "Shalu7Selvaraj";
    int id = 5;
    String name = "Anu";
    int salary = 200000;
    String query = "insert into employee values(?,?,?)";
    Connection con = DriverManager.getConnection(url,
username, password);
    PreparedStatement pst = con.prepareStatement(query);
    pst.setInt(1, id);
    pst.setString(2, name);
```

```

        pst.setInt(3, salary);
        int rows = pst.executeUpdate();
        System.out.println("Number of rows affected:"+rows);
        con.close();
    }

    public static void update() throws Exception{
        String url = "jdbc:mysql://localhost:3306/jdbc";
        String username = "root";
        String password = "Shalu7Selvaraj";
        String query = "update employee set salary = 159000
where id=5";

        Connection con =
DriverManager.getConnection(url,username,password);
        Statement st = con.createStatement();
        int rows = st.executeUpdate(query);
        System.out.println("Number of rows affected:"+rows);
        con.close();
    }

    public static void commitdemo() throws Exception{
        String url = "jdbc:mysql://localhost:3306/jdbc";
        String username = "root";
        String password = "Shalu7Selvaraj";
        String query1 = "update employee set salary = 40000
where id =2";

        String query2 = "update employee set salary =400000
where id = 1";

```

```

        Connection con = DriverManager.getConnection(url,
username, password);
        con.setAutoCommit(false);
        Statement st = con.createStatement();
        int rows1 = st.executeUpdate(query1);
        System.out.println("Rows affected:"+rows1);
        int rows2 = st.executeUpdate(query2);
        System.out.println("Rows affected:"+rows2);
        con.commit();
        con.close();
    }
}

```

TYPE – 2:

DB CONNECTION

```

package JDBC;
import java.sql.*;
public class DatabaseConn {
    public static void main(String [] args) throws Exception{
        String url ="jdbc:mysql://localhost:3306/jdbc";
        String username = "root";
        String password = "Shalu7Selvaraj";
        Connection con = DriverManager.getConnection(url,
username, password);
        System.out.println("connected to DB ");
        con.close();
    }
}

```

```
    }  
}
```

INSERT COMMAND

```
package JDBC;  
import java.sql.*;  
public class InsertType1 {  
    public static void main(String[] args) throws Exception{  
        String url ="jdbc:mysql://localhost:3306/jdbc";  
        String username = "root";  
        String password = "Shalu7Selvaraj";  
        Connection con = DriverManager.getConnection(url,  
username, password);  
        String query ="insert into employee  
values(3,'Abi',350000)";  
        Statement st = con.createStatement();  
        st.executeUpdate(query);  
        System.out.println("Record Inserted Successfully");  
        con.close();  
    }  
}
```

SELECT COMMAND

```
package JDBC;  
import java.sql.*;  
public class SelectCom {
```

```

public static void main(String[] args) throws Exception{
    String url = "jdbc:mysql://localhost:3306/jdbc";
    String username = "root";
    String password ="Shalu7Selvaraj";
    Connection con =
DriverManager.getConnection(url,username,password);
    Statement st = con.createStatement();
    ResultSet rs = st.executeQuery("Select * from employee");
    while (rs.next()) {
        System.out.println(rs.getInt(1)+" "+rs.getString(2)+"
"+rs.getInt(3));
    }
    con.close();
}
}

```

UPDATE COMMAND

```

package JDBC;
import java.sql.*;
public class UpdateCom {
    public static void main(String[] args) throws Exception{
        String url = "jdbc:mysql://localhost:3306/jdbc";
        String username = "root";
        String password ="Shalu7Selvaraj";
        Connection con = DriverManager.getConnection(url,
username, password);

```

```

        String query = "update employee set salary = 250000
where id =2";

        Statement st = con.createStatement();
        st.executeUpdate(query);
        System.out.println("Record Updated Successfully");
        con.close();
    }
}

```

DELETE COMMAND

```

package JDBC;
import java.sql.*;

public class DeleteCom {
    public static void main(String[] args) throws Exception{
        String url = "jdbc:mysql://localhost:3306/jdbc";
        String username = "root";
        String password = "Shalu7Selvaraj";
        Connection con =
DriverManager.getConnection(url,username,password);

        String query ="delete from employee where id =3";
        Statement st = con.createStatement();
        st.executeUpdate(query);
        System.out.println("Record Deleted Successfully");
        con.close();
    }
}

```

TYPE – 3:

```
package JDBC;

import java.sql.*;
import java.util.*;

public class Type3Eg {

    public static void main(String[] args) throws Exception{

        String url = "jdbc:mysql://localhost:3306/jdbc";
        String username = "root";
        String password = "Shalu7Selvaraj";

        Connection con =
DriverManager.getConnection(url,username,password);

        con.setAutoCommit(false);

        System.out.println("Connection Success");

        Statement st = con.createStatement();

        Scanner sc = new Scanner(System.in);

        String createTable = "create table student ("+"name
varchar(20),"+"address varchar(40),"+"city varchar(30))";

        st.executeUpdate(createTable);

        String insertFixed = "insert into student values
('Anu','Navalur','Chennai)";

        st.executeUpdate(insertFixed);

        System.out.println("Hardcode Data Inserted");

        System.out.println("Enter Name: ");

        String name = sc.nextLine();

        System.out.println("Enter Address:");

        String address = sc.nextLine();
```

```

        System.out.println("Enter City:");
        String city = sc.nextLine();
        String insertUser = "insert into student values (?, ?, ?)";
        PreparedStatement ps =
con.prepareStatement(insertUser);
        ps.setString(1, name);
        ps.setString(2, address);
        ps.setString(3, city);
        ps.executeUpdate();
        System.out.println("User Data Inserted");
        String updateQuery = "update student set city = 'bangalore'
where name = 'anu'";
        st.executeUpdate(updateQuery);
        System.out.println("Data Updated");
        String alterQuery = "alter table student add state
varchar(30)";
        st.executeUpdate(alterQuery);
        System.out.println("New Column Added");
        con.commit();
        System.out.println("Transaction Committed");
        String deleteQuery = "delete from student where name
='abi'";
        st.executeUpdate(deleteQuery);
        System.out.println("One Row Deleted");
        ResultSet rs = st.executeQuery("Select * from student");
        System.out.println("\nFinal Table Records");

```

```
        while(rs.next()) {  
            System.out.println(rs.getString(1)+"  
"+rs.getString(2)+" "+rs.getString(3)+" "+rs.getString(4));  
        }  
        rs.close();  
        ps.close();  
        st.close();  
        sc.close();  
        con.close();  
    }  
}
```